Discrete Mathematics

Practice Examples
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Set Theory · Propositional Logic · Counting

Q1 Roster form & set-builder form

PROBLEM

Convert between the two ways of describing a set.

(a) List the set of letters appearing in the word MISSISSIPPI.

(b) Write $\{2, 4, 6, 8, 10\}$ in the form $\{ x \mid P(x) \}$.

(c) Write $\{1, 8, 27, 64, 125\}$ in set-builder form.

(d) Is $\{ x \mid x \text{ is a real number and } x^2 = -1 \}$ empty?

Source · *Kolman §1, Examples 2–4 and Exercises 3, 6–9*

Q1 · Solution

Listing, describing, and the empty set

(a) {M, I, S, P}. The word has 11 letters but the set has only 4 elements — repetition and order carry no information in a set. $|A| = 4$

(b) $\{ x \mid x \text{ is an even positive integer and } x \le 10 \}$, equivalently $\{ 2k \mid k \in \mathbb{Z}^+, 1 \le k \le 5 \}$

(c) $\{ x \mid x = k^3 \text{ for some integer } k \text{ with } 1 \le k \le 5 \}$

(d) Yes. The square of any real number is ≥ 0, so no real number satisfies $x^2 = -1$. The set is $\emptyset$.

Two descriptions, one set

Roster form lists the members. Set-builder form states the admission rule. Both describe the same object; use whichever is shorter.

ANSWER **(a) {M, I, S, P}, $|A| = 4$ (b) even positive integers ≤ 10 (c) cubes of 1–5 (d) $\emptyset$**

Q2 Element vs. subset — the classic trap

PROBLEM

Let $A = \{1, \{2, 3\}, 4\}$. Mark each statement true or false, and justify.

(a) $3 \in A$ (b) $\{1, 4\} \subseteq A$ (c) $\{2, 3\} \subseteq A$

(d) $\{2, 3\} \in A$ (e) $\{4\} \in A$ (f) $\{1, 2, 3\} \subseteq A$

(g) What are $|A|$ and $|P(A)|$?

Source · *Kolman §1, Exercise 5. Read A as having exactly three members.*

Q2 · Solution

A has exactly three elements: 1, the set {2,3}, and 4

(a) FALSE. 3 lives inside the element {2, 3}. It is not itself a member of A.

(b) TRUE. Both 1 and 4 are members of A, so every element of {1, 4} is in A.

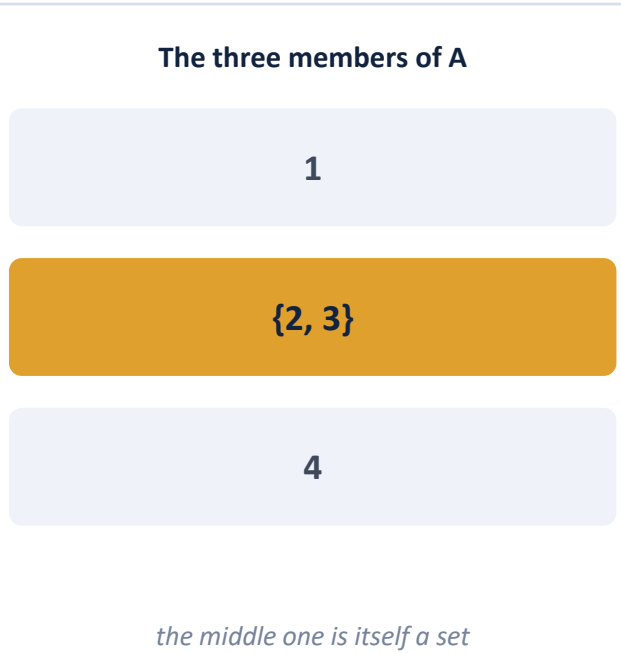
(c) FALSE. For {2,3} ⊆ A we would need 2 ∈ A and 3 ∈ A. Neither holds.

(d) TRUE. The set {2, 3} is literally listed as one of A's three members.

(e) FALSE. {4} is a subset of A, not an element of it. 4 ∈ A but {4} ∉ A.

(f) FALSE. Same reason as (c) — 2 and 3 are not members of A.

(g) |A| = 3, so |P(A)| = 2^3 = 8.



ANSWER **F, T, F, T, F, F · |A| = 3, |P(A)| = 8**

Q3 Set operations — computation drill

PROBLEM

Let $U = \{a, b, c, d, e, f, g, h, k\}$, $A = \{a, b, c, g\}$, $B = \{d, e, f, g\}$, $C = \{a, c, f\}$, $D = \{f, h, k\}$.

Compute: (a) $A \cup B$ (b) $B \cup C$ (c) $A \cap C$ (d) $B \cap D$
(e) $(A \cup B) - C$ (f) $A - B$ (g) $\bar{A}$ (h) $A \oplus B$

Source · *Kolman §2, Exercises 1–2. $A \oplus B = (A - B) \cup (B - A)$*

Q3 · Solution

Work strictly from the definitions

(a) $A \cup B = \{a, b, c, d, e, f, g\}$ — everything in either set, g listed once

(b) $B \cup C = \{a, c, d, e, f, g\}$

(c) $A \cap C = \{a, c\}$ — only the members common to both

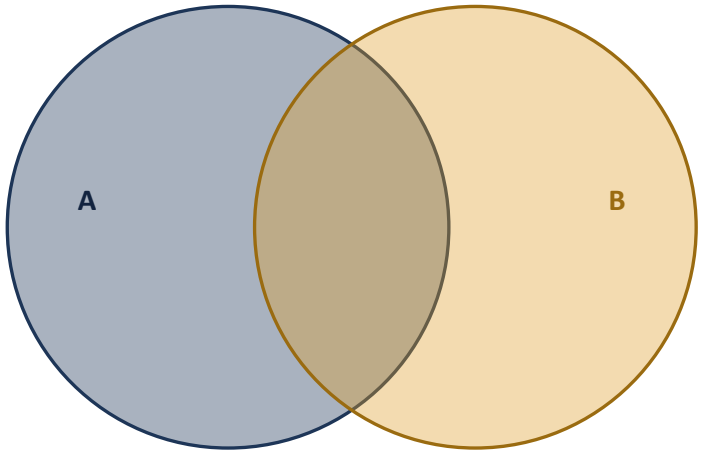
(d) $B \cap D = \{f\}$

(e) $A \cup B = \{a,b,c,d,e,f,g\}$; now delete the members of $C = \{a,c,f\}$ → $\{b, d, e, g\}$

(f) $A - B = \{a, b, c\}$ — in A but not in B; g is removed

(g) $\bar{A} = U - A = \{d, e, f, h, k\}$

(h) $A \oplus B = (A-B) \cup (B-A) = \{a,b,c\} \cup \{d,e,f\} = \{a, b, c, d, e, f\}$ — g is excluded, it is in both



$A \oplus B$ shades everything except the overlap

ANSWER **(h) $A \oplus B = \{a, b, c, d, e, f\}$ — note g disappears**

Q4 · Solution 1/2

Part 1 — numerical verification

$A \cup B = \{1, 2, 4, 5, 6, 8, 9\}$

$(A \cup B)' = U - (A \cup B) = \{3, 7\}$

$\bar{A} = \{3, 5, 7, 9\}$ and $\bar{B} = \{1, 3, 6, 7, 8\}$

$\bar{A} \cap \bar{B} = \{3, 7\}$

The two sides agree. But one example is not a proof — it only tells us the claim is worth proving.

ANSWER **Both sides equal {3, 7} ✓**

Q4 · Solution 2/2

Part 2 — general proof by double inclusion

To prove two sets equal, show each is a subset of the other.

Step 1: show $(A \cup B)' \subseteq \bar{A} \cap \bar{B}$

Let $x \in (A \cup B)'$. — *assume a generic element*
Then $x \notin A \cup B$. — *definition of complement*
So $x \notin A$ and $x \notin B$. — *definition of union*
Hence $x \in \bar{A}$ and $x \in \bar{B}$, so $x \in \bar{A} \cap \bar{B}$. — *definition of ∩*

Step 2: show $\bar{A} \cap \bar{B} \subseteq (A \cup B)'$

Let $x \in \bar{A} \cap \bar{B}$, so $x \notin A$ and $x \notin B$. — *definition of ∩*
Then x belongs to neither set, so $x \notin A \cup B$. — *definition of union*
Hence $x \in (A \cup B)'$. — *definition of complement*
Both inclusions hold, therefore the sets are equal. ■

ANSWER **$(A \cup B)' = \bar{A} \cap \bar{B}$ — proved by double inclusion**

Q5 Power sets

PROBLEM

Power sets, edge cases, and a counting argument.

(a) List every element of $P(A)$ for $A = \{1, 2, 3\}$.

(b) What is $P(\emptyset)$? (c) What is $P(P(\emptyset))$?

(d) Prove that if $|A| = n$ then $|P(A)| = 2^n$.

Source · *Kolman §1, Example 11 and Counting §1, Example 2*

Q5 · Solution

Counting subsets with a bit-string bijection

(a) $\emptyset$, {1}, {2}, {3}, {1,2}, {1,3}, {2,3}, {1,2,3} — eight subsets, i.e. 2^3

(b) $P(\emptyset) = \{ \emptyset \}$. Note carefully: $|\emptyset| = 0$ but $|P(\emptyset)| = 1$. The empty set has exactly one subset — itself.

(c) $P(\{\emptyset\}) = \{ \emptyset, \{\emptyset\} \}$, which has 2 elements.

(d) The proof

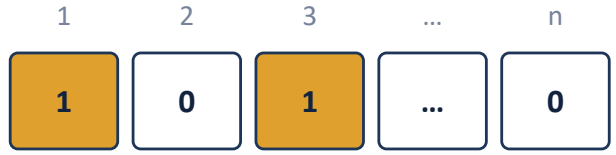
Fix an ordering of A as $a_1, a_2, \ldots, a_n$.

Any subset $S \subseteq A$ is completely described by n yes/no answers: is a_i in S ?

Record the answers as a bit string of length n — 1 for 'in', 0 for 'out'.

This pairing is one-to-one: different subsets give different strings, and every string names a subset.

Each of the n positions can be filled in 2 ways, independently, so by the product rule there are $2 \cdot 2 \cdot \ldots \cdot 2 = 2^n$ strings, hence 2^n subsets. ■



1 = in the subset, 0 = out

{1,3} ↔ 101

ANSWER $|P(A)| = 2^n$ · $P(\emptyset) = \{\emptyset\}$, and $\emptyset \neq \{\emptyset\}$

Q6 Proving an identity with set laws

PROBLEM

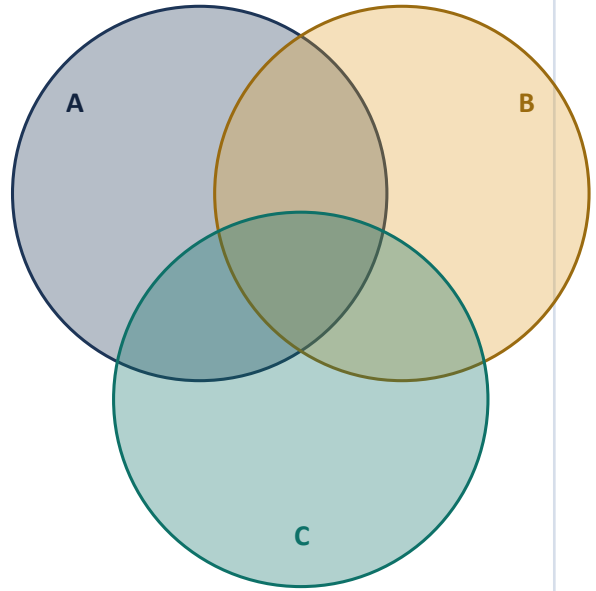
Prove that $A - (B \cup C) = (A - B) \cap (A - C)$, using the algebraic laws of set operations rather than element-chasing. Justify every line.

Source · *Start by rewriting difference as intersection with a complement.*

Q6 · Solution

An algebraic proof — every line names its law

$A - (B \cup C)$ — *start*
$= A \cap (B \cup C)'$ — *definition of difference*
$= A \cap (B' \cap C')$ — *De Morgan*
$= (A \cap A) \cap (B' \cap C')$ — *idempotent law: $A = A \cap A$*
$= (A \cap B') \cap (A \cap C')$ — *commutative + associative*
$= (A - B) \cap (A - C)$ — *definition of difference* ■



A minus everything B or C covers

ANSWER $A - (B \cup C) = (A - B) \cap (A - C)$

Q7 Inclusion–exclusion for two sets

PROBLEM

Two applications of $|A \cup B| = |A| + |B| - |A \cap B|$.

(a) $A = \{a, b, c, d, e\}$, $B = \{c, e, f, h, k, m\}$. Verify the formula.

(b) A company wants to hire 25 systems programmers and 40 applications programmers. Ten of those hired will be expected to do both jobs. How many programmers must be hired?

Source · *Kolman §2, Theorem 2 and Examples 7, 9*

Q7 · Solution

Why we subtract the overlap

(a) Verification

$A \cup B = \{a, b, c, d, e, f, h, k, m\}$, so $|A \cup B| = 9$

$A \cap B = \{c, e\}$, so $|A \cap B| = 2$; also $|A| = 5$, $|B| = 6$

$|A| + |B| - |A \cap B| = 5 + 6 - 2 = 9$ ✓

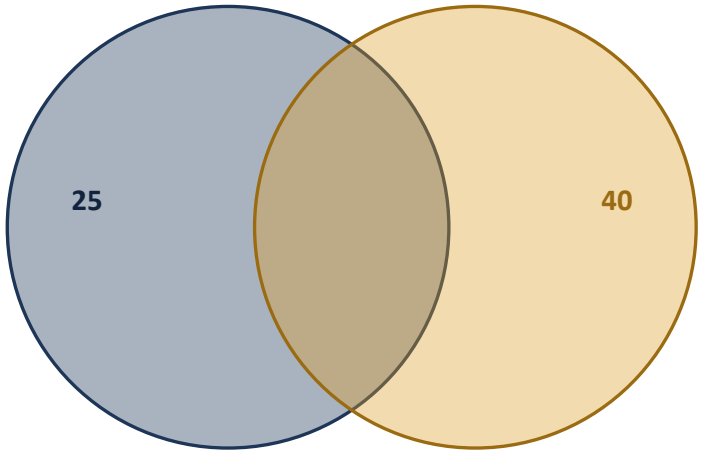
(b) The hiring problem

Let A = systems programmers hired, B = applications programmers hired.

Given: $|A| = 25$, $|B| = 40$, $|A \cap B| = 10$. We want $|A \cup B|$ — the total headcount.

$|A \cup B| = 25 + 40 - 10 = 55$

So 55 people must be hired, not 65.



the 10 in the middle were counted twice

ANSWER **(a) both sides = 9 (b) 55 programmers**

Q8 Inclusion–exclusion for three sets

PROBLEM

A commuter survey allowed multiple answers. Results: BUS 30, TRAIN 35, AUTOMOBILE 100, BUS and TRAIN 15, BUS and AUTO 15, TRAIN and AUTO 20, all three 5.

(a) How many people completed a form?

(b) How many use exactly one method?

(c) How many use the bus but not the train?

Source · *Kolman §2, Theorem 3 and Example 10. Fill the Venn diagram from the centre outwards.*

Q8 · Solution 1/2

Part 1 — the total

$$|B \cup T \cup A| = |B| + |T| + |A| - |B \cap T| - |B \cap A| - |T \cap A| + |B \cap T \cap A|$$

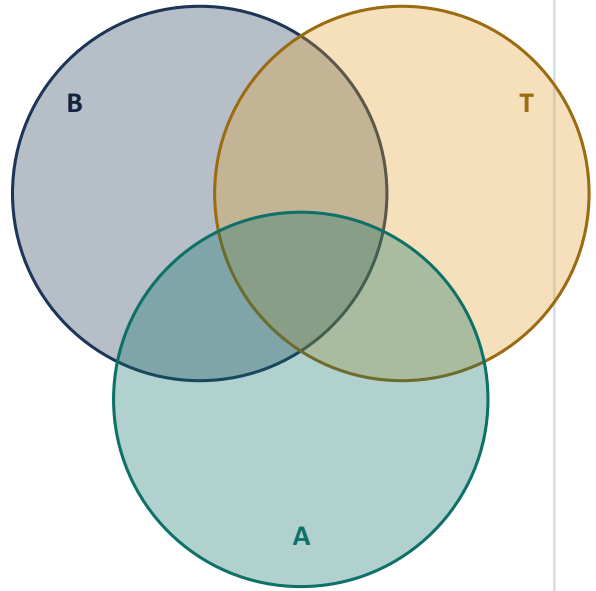
$$= 30 + 35 + 100 - 15 - 15 - 20 + 5$$

$$= 165 - 50 + 5 = 120$$

So 120 people responded.

Why the last term is added back

The three people-in-all-three got counted 3 times in the first bracket, then subtracted 3 times in the second. That leaves them at zero — so we add them back once.



add singles, subtract pairs, add back the triple

ANSWER **120 respondents**

Q8 · Solution 2/2

Part 2 — filling the seven regions

Always start at the centre and work outwards.

All three: 5

B ∩ T only = 15 − 5 = 10

B ∩ A only = 15 − 5 = 10

T ∩ A only = 20 − 5 = 15

Bus only = 30 − 10 − 10 − 5 = 5

Train only = 35 − 10 − 15 − 5 = 5

Auto only = 100 − 10 − 15 − 5 = 70

(b) exactly one method = 5 + 5 + 70 = 80

(c) bus but not train = |B| − |B ∩ T| = 30 − 15 = 15

Check: 5+5+70+10+10+15+5 = 120 ✓ matches Part 1

ANSWER **(b) 80 people use exactly one method (c) 15 use bus but not train**

Q9 Inclusion–exclusion run backwards

PROBLEM

In a class of 60 students, 35 like Java, 30 like Python, and 12 like neither.

(a) How many like both languages?

(b) How many like exactly one language?

Source · *The word 'neither' places people outside the union, not inside it.*

Q9 · Solution

The unknown is inside the formula, not outside it

The 12 who like neither sit outside J ∪ P, in the universal set.

$|J \cup P| = |U| - |\text{neither}| = 60 - 12 = 48$

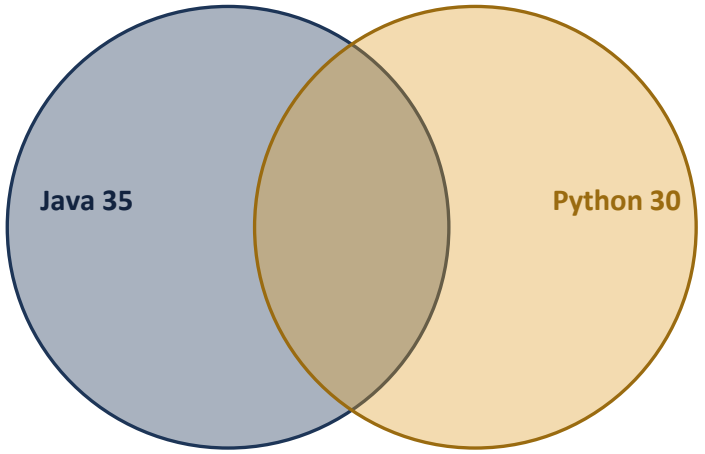
Now apply the formula with the union known and the intersection unknown:

$48 = 35 + 30 - |J \cap P|$ — *inclusion–exclusion*

$|J \cap P| = 65 - 48 = 17$ — *rearrange*

(b) exactly one = $|J \cup P| - |J \cap P| = 48 - 17 = 31$

Cross-check: $(35 - 17) + (30 - 17) = 18 + 13 = 31$ ✓



12 sit outside both circles

ANSWER **(a) 17 like both (b) 31 like exactly one**

Q10 Proof by contraposition

PROBLEM

Let n be an integer. Prove that if n^2 is odd, then n is odd.

Source · *Kolman, Methods of Proof, Example 6. Try the direct proof first and notice where it stalls.*

Q10 · Solution

Choose the direction that hands you an equation

Why not a direct proof?

A direct proof starts from 'n^2 is odd', i.e. $n^2 = 2k + 1$. From that you cannot easily extract anything about n — there is no clean square root to take. The hypothesis is hard to use.

Switch to the contrapositive

$p \Rightarrow q$ is logically equivalent to $\neg q \Rightarrow \neg p$, so instead prove: if n is even, then n^2 is even.

Assume n is even. *— this is ¬q*
Then $n = 2k$ for some integer k. *— definition of even — an equation we can use*
$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ *— algebra*
Since $2k^2$ is an integer, n^2 is even. *— this is ¬p*
We have proved $\neg q \Rightarrow \neg p$, hence $p \Rightarrow q$. ■

ANSWER **Proved: n^2 odd ⇒ n odd, via the contrapositive**

Q11 Proof by contradiction

PROBLEM

Prove that √2 is irrational — that is, prove there is no rational number p/q whose square is 2.

Source · *Kolman, Methods of Proof, Example 7. Begin by assuming the opposite, in lowest terms.*

Q11 · Solution

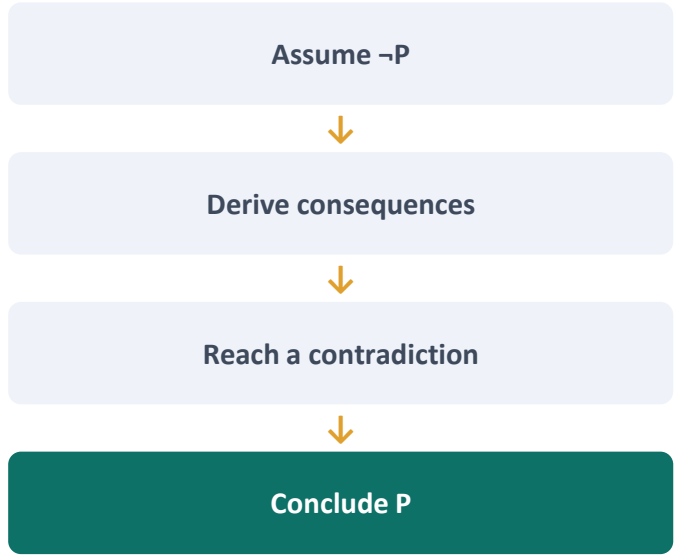
Assume the opposite and derive an absurdity

Suppose, for contradiction, that $\sqrt{2}$ is rational.

Then $\sqrt{2} = p/q$ for integers p, q with $q \neq 0$, in lowest terms — that is, $\gcd(p, q) = 1$. *— we may always reduce the fraction*
Squaring: $2 = p^2/q^2$, hence $p^2 = 2q^2$. *— algebra*
So p^2 is even, and therefore p is even. *— by the result of Q10*
Write $p = 2n$ for some integer n.

Then $(2n)^2 = 2q^2$, so $4n^2 = 2q^2$, giving $q^2 = 2n^2$. *— substitute and simplify*
So q^2 is even, and therefore q is even. *— Q10 again*
But now 2 divides both p and q, contradicting $\gcd(p, q) = 1$.

The assumption is therefore false: $\sqrt{2}$ is irrational. ■



shape of every proof by contradiction

ANSWER **$\sqrt{2}$ is irrational — the assumption of rationality forces a contradiction**

Q12 Proof by contradiction — infinitude of primes

PROBLEM

Prove that there are infinitely many prime numbers.

Source · *Assume the list of primes is finite and complete, then build a number that breaks it.*

Q12 · Solution

Euclid's argument

Suppose, for contradiction, that there are only finitely many primes: $p_1, p_2, \ldots, p_n$, and that this list is complete.

Define $N = (p_1 \cdot p_2 \cdot \ldots \cdot p_n) + 1$.

$N > 1$, so N has at least one prime divisor p. *— every integer > 1 has a prime factor*
Since the list is complete, $p = p_i$ for some i.

Then p_i divides N, and p_i also divides the product $p_1 \cdots p_n$.

So p_i divides their difference: $N - (p_1 \cdots p_n) = 1$.

But no prime divides 1, since every prime is ≥ 2. Contradiction.

Hence no finite list can contain all the primes. ■

ANSWER **There are infinitely many primes**

Q13 Mathematical induction — a summation

PROBLEM

Prove by mathematical induction that $1 + 2 + 3 + \ldots + n = n(n+1)/2$ for every integer $n \geq 1$.

Source · *Kolman, Mathematical Induction, Example 1. Write out P(n) before you start.*

Q13 · Solution

The four-part ritual: statement, basis, hypothesis, step

Statement

Let P(n) be the claim: $1 + 2 + \dots + n = n(n+1)/2$. Here $n_0 = 1$.

Basis step — verify P(1)

LHS = 1. RHS = 1(1+1)/2 = 1. Equal, so P(1) holds. ✓

Induction hypothesis

Assume P(k) is true for some fixed $k \geq 1$: $1 + 2 + \dots + k = k(k+1)/2$.

Induction step — show P(k+1) follows

$1 + 2 + \dots + k + (k+1)$ — *left side of P(k+1)*
$= k(k+1)/2 + (k+1)$ — *substitute the hypothesis*
$= (k+1) \cdot [\, k/2 + 1 \,]$ — *factor out (k+1)*
$= (k+1)(k+2)/2$ — *combine the bracket*
$= (k+1)((k+1)+1)/2$ — *right side of P(k+1)* ✓

P(1) holds and P(k) ⇒ P(k+1), so by the principle of mathematical induction P(n) holds for all $n \geq 1$. ■



ANSWER **$1 + 2 + \dots + n = n(n+1)/2$ for all $n \geq 1$**

Q14 Induction — divisibility

PROBLEM

Prove by induction that 3 divides $n^3 + 2n$ for every integer $n \geq 1$.

Source · *State P(n) as 'n³ + 2n = 3m for some integer m' — divisibility becomes an equation.*

Q14 · Solution

Extract the hypothesis surgically from the expansion

Statement

P(n): $n^3 + 2n = 3m$ for some integer m.

Basis

n = 1: $1 + 2 = 3 = 3(1)$. ✓

Hypothesis

Assume $k^3 + 2k = 3m$ for some integer m, where $k \geq 1$.

Step — examine $(k+1)^3 + 2(k+1)$

$= k^3 + 3k^2 + 3k + 1 + 2k + 2$ *— expand everything*
$= (k^3 + 2k) + 3k^2 + 3k + 3$ *— regroup to expose P(k)*
$= 3m + 3(k^2 + k + 1)$ *— substitute the hypothesis*
$= 3(m + k^2 + k + 1)$ *— factor out 3*
Since $m + k^2 + k + 1$ is an integer, 3 divides $(k+1)^3 + 2(k+1)$.

By induction, $3 \mid (n^3 + 2n)$ for all $n \geq 1$. ■

ANSWER **3 divides $n^3 + 2n$ for every $n \geq 1$**

Q15 Induction — an inequality with a shifted basis

PROBLEM

Prove that $2^n > n^2$ for every integer $n \geq 5$. Explain why the basis cannot be taken at $n = 1$.

Source · *Check n = 1, 2, 3, 4 first. Some of them fail — that is the point.*

Q15 · Solution

The starting rung is not always the first one

Why $n_0 = 5$

n = 2: 4 vs 4 — not strictly greater. n = 3: 8 vs 9 — false. n = 4: 16 vs 16 — not strictly greater.

n = 5: 32 vs 25 — true. So the claim genuinely begins at 5.

Basis

P(5): $2^5 = 32 > 25 = 5^2$. ✓

Hypothesis

Assume $2^k > k^2$ for some fixed $k \geq 5$.

Step

$2^{k+1} = 2 \cdot 2^k > 2k^2$ *— by the hypothesis*

Now it suffices to show $2k^2 \geq (k+1)^2$ for $k \geq 5$:

$2k^2 \geq k^2 + 2k + 1 \iff k^2 - 2k - 1 \geq 0 \iff (k-1)^2 \geq 2$ *— complete the square*

For $k \geq 5$ we have $(k-1)^2 \geq 16 \geq 2$, so the claim holds.

Chaining: $2^{k+1} > 2k^2 \geq (k+1)^2$. Hence P(k+1).

By induction, $2^n > n^2$ for all $n \geq 5$. ■

ANSWER **$2^n > n^2$ for all $n \geq 5$ — and it is false for n = 3 and n = 4**

Q16 Induction on sets — generalised De Morgan

PROBLEM

Let $A_1, A_2, \ldots, A_n$ be any n sets. Prove by induction that $\overline{(A_1 \cup A_2 \cup \ldots \cup A_n)} = \overline{A_1} \cap \overline{A_2} \cap \ldots \cap \overline{A_n}$.

Source · *Kolman, Mathematical Induction, Example 2. The two-set law from Q4 is the engine of the step.*

Q16 · Solution

Induction is not only for numbers

Statement

P(n): the identity holds for any collection of n sets.

Basis

P(1) reads $\overline{A_1} = \overline{A_1}$, which is trivially true. ✓

(P(2) is the two-set De Morgan law proved in Q4 — we take it as known.)

Hypothesis

Assume P(k): $(A_1 \cup \ldots \cup A_k)' = \overline{A_1} \cap \ldots \cap \overline{A_k}$.

Step

$(A_1 \cup \ldots \cup A_k \cup A_{k+1})'$ — *left side of P(k+1)*
$= ((A_1 \cup \ldots \cup A_k) \cup A_{k+1})'$ — *associativity of ∪ — bracket the first k*
$= (A_1 \cup \ldots \cup A_k)' \cap \overline{A_{k+1}}$ — *two-set De Morgan*
$= (\overline{A_1} \cap \ldots \cap \overline{A_k}) \cap \overline{A_{k+1}}$ — *induction hypothesis*
$= \overline{A_1} \cap \ldots \cap \overline{A_{k+1}}$ — *right side of P(k+1)* ✓
By induction the identity holds for all $n \geq 1$. ■

ANSWER **Complement of a union of n sets = intersection of the n complements**

UNIT 2

Propositional Logic

Propositions, truth tables, logical equivalences, translation, and valid argument

Propositions | Truth tables | Equivalences | English → symbols | Rules of

12 solved problems

Q17 What counts as a proposition?

PROBLEM

Which of the following are propositions? Give the truth value where you can.

(a) The earth is round. (b) $2 + 3 = 5$ (c) Do you speak English?

(d) $3 - x = 5$ (e) Take two aspirins.

(f) The surface temperature of Venus is 800°F.

(g) The sun will come out tomorrow. (h) This sentence is false.

Source · *Kolman, Logic §1, Example 1. The test is: declarative, and exactly one truth value.*

Q17 · Solution

Declarative, and exactly one truth value — both conditions

(a) Proposition. True.

(b) Proposition. True.

(c) Not a proposition — it is a question, not a declaration.

(d) Not a proposition — its truth depends on x. This is an open sentence, or predicate. It becomes a proposition once x is fixed, or once it is quantified.

(e) Not a proposition — it is a command.

(f) Proposition. We may not know the value, but the sentence has one. Ignorance is not the same as indeterminacy.

(g) Proposition. Its value will be settled tomorrow; that is enough.

(h) Not a proposition — assigning it True forces False and vice versa. It is a paradox, not a statement.

ANSWER **Propositions: (a), (b), (f), (g). Not: (c), (d), (e), (h)**

Q18 Truth tables and classification

PROBLEM

Construct the truth table for $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$. Classify it. Then classify $p \wedge \neg p$ and $(p \Rightarrow q) \wedge (p \vee q)$.

Source · *Kolman, Logic §2, Examples 4–5. Kolman writes ~p for ¬p.*

Q18 · Solution

Build inner columns first, then combine

The main table

Every row of the final column is T, so the statement is a TAUTOLOGY.

This tautology has a name: it is the law of contraposition. It is the licence for the proof technique used in Q10.

p ∧ ¬p

p = T gives T ∧ F = F; p = F gives F ∧ T = F. Always false — a CONTRADICTION.

(p ⇒ q) ∧ (p ∨ q)

TT: T ∧ T = T TF: F ∧ T = F FT: T ∧ T = T FF: T ∧ F = F

Both T and F occur, so this is a CONTINGENCY.

p	q	p⇒q	¬q	¬p	¬q⇒¬p	⇔
T	T	T	F	F	T	**T**
T	F	F	T	F	F	**T**
F	T	T	F	T	T	**T**
F	F	T	T	T	T	**T**

build inner columns first

ANSWER **Tautology · Contradiction · Contingency — the three possible verdicts**

Q19 Converse, inverse, contrapositive

PROBLEM

For each implication below, write the converse, inverse, and contrapositive, and say which are logically equivalent to the original.

(a) If it is raining, then I get wet.

(b) If a program compiles, then it has no syntax errors.

Source · *Kolman, Logic §2, Example 2*

Q19 · Solution

Only one of the three is equivalent

(a) p: it is raining. q: I get wet.

Original p ⇒ q — If it is raining, then I get wet.

Converse q ⇒ p — If I get wet, then it is raining.

Inverse ¬p ⇒ ¬q — If it is not raining, then I do not get wet.

Contrapositive ¬q ⇒ ¬p — If I do not get wet, then it is not raining.

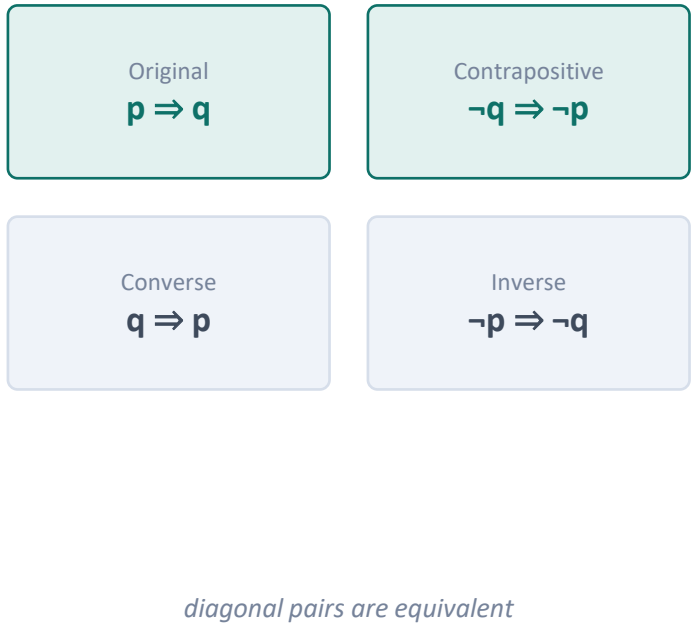
Equivalences: original ≡ contrapositive, and converse ≡ inverse. The original is NOT equivalent to its converse.

(b) A computing example that settles the argument

Original: if it compiles, it has no syntax errors — true.

Converse: if it has no syntax errors, it compiles — FALSE. It could still fail on a missing header, a link error, a type error.

Contrapositive: if it has a syntax error, it does not compile — true, and says exactly what the original says.



ANSWER original ≡ contrapositive · converse ≡ inverse · original ≢ converse

Q20 Rewriting and negating implications

PROBLEM

Show that $p \Rightarrow q \equiv \neg p \vee q$ by truth table. Then use it to find a simplified form of $\neg(p \Rightarrow q)$.

Source · *This single equivalence removes ⇒ from any formula. It is the most useful identity in the unit.*

Q20 · Solution

Implication is really a disjunction in disguise

The truth table columns for p ⇒ q and ¬p ∨ q agree in all four rows, so the two are logically equivalent.

Now negate it

¬(p ⇒ q) *— start*
≡ ¬(¬p ∨ q) *— rewrite the implication*
≡ ¬(¬p) ∧ ¬q *— De Morgan*
≡ p ∧ ¬q *— double negation*

Read the result in English

The negation of 'if it rains, I get wet' is NOT another if-then. It is: 'it rained AND I stayed dry.' The only way an implication fails is a true hypothesis with a false conclusion — exactly one row of the table.

p	q	p⇒q	¬p	¬p∨q
T	T	T	F	**T**
T	F	F	F	**F**
F	T	T	T	**T**
F	F	T	T	**T**

build inner columns first

ANSWER **p ⇒ q ≡ ¬p ∨ q** **and** **¬(p ⇒ q) ≡ p ∧ ¬q**

Q21 Simplification using logical laws

PROBLEM

Show that $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$, using logical equivalences only. Name the law at every step.

Source · *A truth table would settle it in four rows — but the point here is algebraic fluency.*

Q21 · Solution

Every line carries its licence

¬(p ∨ (¬p ∧ q)) — *start*
≡ ¬p ∧ ¬(¬p ∧ q) — *De Morgan*
≡ ¬p ∧ (p ∨ ¬q) — *De Morgan + double negation*
≡ (¬p ∧ p) ∨ (¬p ∧ ¬q) — *distributive law*
≡ F ∨ (¬p ∧ ¬q) — *negation law: ¬p ∧ p ≡ F*
≡ ¬p ∧ ¬q — *identity law: F ∨ X ≡ X* ■

Laws used

De Morgan
¬(p∧q) ≡ ¬p∨¬q

Distributive
p∧(q∨r) ≡ (p∧q)∨(p∧r)

Negation
p∧¬p ≡ F

Identity
F∨p ≡ p

Double neg.
¬(¬p) ≡ p

ANSWER **¬(p ∨ (¬p ∧ q)) ≡ ¬p ∧ ¬q**

Q22 Translating English into logic — the hard connectives

PROBLEM

Translate into symbolic form.

(a) Let a: 'you can access the internet from campus', c: 'you are a computer science major', f: 'you are a freshman'. Translate: You can access the internet from campus ONLY IF you are a computer science major or you are not a freshman.

(b) Let r: 'you can ride the roller coaster', u: 'you are under 4 feet tall', o: 'you are older than 16'. Translate: You cannot ride the roller coaster if you are under 4 feet tall UNLESS you are older than 16.

Source · *'only if' and 'unless' are the two that decide the marks.*

Q22 · Solution

Where the clause lands is the whole question

(a)

'A only if B' translates to $A \Rightarrow B$. The clause after 'only if' states a NECESSARY condition, and necessary conditions go on the right of ⇒.

Answer: $a \Rightarrow (c \vee \neg f)$

Common wrong answer: $(c \vee \neg f) \Rightarrow a$. That says being a CS major guarantees access, which is not what was claimed.

(b)

'unless' means 'if not'. So 'X unless Y' becomes $\neg Y \Rightarrow X$.

The sentence says: being under 4 feet blocks you — unless you are over 16.

Answer: $(u \wedge \neg o) \Rightarrow \neg r$

The card to memorise

p if q → $q \Rightarrow p$ p only if q → $p \Rightarrow q$ p unless q → $\neg q \Rightarrow p$

p is sufficient for q → $p \Rightarrow q$ p is necessary for q → $q \Rightarrow p$

ANSWER **(a) $a \Rightarrow (c \vee \neg f)$ (b) $(u \wedge \neg o) \Rightarrow \neg r$**

Q23 Translation drill

PROBLEM

Let p: 'it is below freezing', q: 'it is snowing', r: 'the roads are cleared'. Express each sentence in symbols.

(a) It is below freezing and snowing.

(b) It is below freezing but not snowing.

(c) It is neither below freezing nor snowing.

(d) If it is below freezing, it is also snowing.

(e) It is either below freezing or snowing, but it is not snowing if it is below freezing.

(f) The roads are cleared only if it is not snowing.

(g) It is necessary that it be below freezing for it to be snowing.

Q23 · Solution

Strip the decoration, keep the logic

(a) $p \wedge q$

(b) $p \wedge \neg q$ — 'but' is just $\wedge$; it carries contrast in English but no logical content

(c) $\neg p \wedge \neg q$ — 'neither X nor Y' is always $\neg X \wedge \neg Y$

(d) $p \Rightarrow q$

(e) $(p \vee q) \wedge (p \Rightarrow \neg q)$ — two claims joined by 'but', so conjoin them

(f) $r \Rightarrow \neg q$ — 'only if' puts its clause on the right

(g) $q \Rightarrow p$ — 'p is necessary for q' means q cannot happen without p

ANSWER **(b) 'but' = $\wedge$ (c) neither/nor = $\neg p \wedge \neg q$ (g) necessary $\Rightarrow$ arrow points at p**

Q24 Negating compound statements

PROBLEM

Write the negation of each statement, first in symbols and then in plain English.

(a) The processor is fast and the memory is cheap.

(b) If I study, then I will pass.

(c) The file is either read-only or hidden.

Q24 · Solution

Negation flips the connective, not just the parts

(a) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

English: 'The processor is not fast, OR the memory is not cheap.'

Note the AND became an OR. Denying that both hold only requires that at least one fails.

(b) $\neg(p \Rightarrow q) \equiv p \wedge \neg q$

English: 'I study and I do not pass.'

The negation of an if-then is NOT an if-then. It is the single failing case.

(c) $\neg(p \vee q) \equiv \neg p \wedge \neg q$

English: 'The file is neither read-only nor hidden.'

The OR became an AND.

Negation summary

¬ over ∧
$\neg p \vee \neg q$

¬ over ∨
$\neg p \wedge \neg q$

¬ over ⇒
$p \wedge \neg q$

ANSWER $\neg(p \wedge q) \equiv \neg p \vee \neg q$ · $\neg(p \vee q) \equiv \neg p \wedge \neg q$ · $\neg(p \Rightarrow q) \equiv p \wedge \neg q$

Q25 Consistency of system specifications

PROBLEM

A system's specifications state: (1) The diagnostic message is stored in the buffer or it is retransmitted. (2) The diagnostic message is not stored in the buffer. (3) If the diagnostic message is stored in the buffer, then it is retransmitted.

(a) Are these three specifications consistent?

(b) What happens if we also require: the diagnostic message is not retransmitted?

Source · *Consistent means: there exists at least one truth assignment satisfying every specification.*

Q25 · Solution

Consistency = satisfiability

Let b: the message is stored in the buffer, r: the message is retransmitted.

Specifications: (1) $b \vee r$ (2) $\neg b$ (3) $b \Rightarrow r$

(a) Search for a satisfying assignment

Specification (2) forces b = F.

With b = F, specification (3) is vacuously true whatever r is.

With b = F, specification (1) requires r = T.

Take b = F, r = T: all three specifications hold. The set is CONSISTENT.

(b) Add specification (4): $\neg r$

Now r = F. Combined with b = F from (2), specification (1) $b \vee r$ becomes $F \vee F = F$.

Specification (1) fails, and no other assignment is available. The enlarged set is INCONSISTENT.

ANSWER **(a) consistent, via b = F, r = T (b) adding $\neg r$ makes them inconsistent**

Q26 Valid, invalid, and the difference from true

PROBLEM

Decide whether each argument is valid. Name the form.

(a) If taxes are lowered, income rises. Income rises. ∴ Taxes were lowered.

(b) If you invest in the stock market, you get rich. If you get rich, you are happy. ∴ If you invest in the stock market, you are happy.

(c) Smoking is healthy. If smoking is healthy, cigarettes are prescribed by physicians. ∴ Cigarettes are prescribed by physicians.

Source · *Kolman, Methods of Proof, Examples 2, 4, 5*

Q26 · Solution

Validity is about shape, not about truth

(a) $p \Rightarrow q,\ q\ \therefore p$

INVALID — the fallacy of affirming the consequent.

Counterexample row: p = F, q = T. Both premises are true, the conclusion is false. One such row is enough to destroy validity.

(b) $p \Rightarrow q,\ q \Rightarrow r\ \therefore p \Rightarrow r$

VALID — hypothetical syllogism. The implication $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$ is a tautology.

(c) $p,\ p \Rightarrow q\ \therefore q$

VALID — modus ponens. And yet the conclusion is plainly false.

There is no contradiction. The argument is valid but not SOUND, because the premise 'smoking is healthy' is false.

The distinction

VALID: if the premises are true, the conclusion must be true.

SOUND: valid, AND the premises actually are true.

ANSWER **(a) invalid (b) valid (c) valid but not sound**

Q27 Building an argument — proof by cases

PROBLEM

From the premises (1) $p \vee q$, (2) $p \Rightarrow r$, (3) $q \Rightarrow r$, derive the conclusion r. Justify each step.

Source · *Two of the premises point to the same place. That is the shape of a case analysis.*

Q27 · Solution

Constructive dilemma — the engine behind every proof by cases

Premise (1) tells us at least one of p, q holds. We do not know which — so consider both possibilities.

Case 1: p is true

From (2) $p \Rightarrow r$ and modus ponens, r follows.

Case 2: q is true

From (3) $q \Rightarrow r$ and modus ponens, r follows.

Combine

Premise (1) guarantees at least one case occurs, and both cases deliver r. Therefore r holds. $\therefore r$ ■

Equivalent check

The implication $((p \vee q) \wedge (p \Rightarrow r) \wedge (q \Rightarrow r)) \Rightarrow r$ is a tautology; an eight-row truth table confirms it.

ANSWER **r follows — by constructive dilemma (proof by cases)**

Q28 A logic puzzle — knights and knaves

PROBLEM

On an island every inhabitant is either a knight, who always tells the truth, or a knave, who always lies. You meet A and B. A says: “I am a knave, or B is a knight.” What are A and B?

Source · *Model the island rule as an equivalence, not as an implication.*

Q28 · Solution

One modelling insight, then two rows of checking

Let a: 'A is a knight' and b: 'B is a knight'.

A's statement is $S = \neg a \vee b$.

The island rule, stated precisely

A knight says only true things and a knave says only false things, so: $a \Leftrightarrow S$.

That is, A is a knight exactly when what A said is true. This single line is the whole modelling step.

Case a = T

$S = \neg T \vee b = F \vee b = b$. The rule $a \Leftrightarrow S$ becomes $T \Leftrightarrow b$, forcing $b = T$. Consistent.

Case a = F

$S = \neg F \vee b = T \vee b = T$. The rule becomes $F \Leftrightarrow T$, which is false. Contradiction — case rejected.

Only the first case survives.

Sanity check: A is a knight, so A's statement must be true. 'I am a knave or B is a knight' = $F \vee T = T$. ✓

ANSWER **Both A and B are knights**

UNIT 3

Counting

The basics of counting, sum and product rules, permutations and combinations

Sum & product rules | Permutations | Repeated objects | Combinations

14 solved problems

Q29 Sum rule versus product rule

PROBLEM

There are 3 routes from Atlanta to Athens, 4 routes from Athens to Augusta, and 2 direct routes from Atlanta to Augusta.

(a) How many ways are there to travel from Atlanta to Augusta?

(b) How many ways to make the round trip Atlanta → Augusta → Atlanta?

(c) Same as (b), but no route may be used twice. [stretch]

Source · *Kolman, Counting §1, Exercises 35–36. Decide first whether the situation is 'or' or 'and'.*

Q29 · Solution 1/2

Alternatives add; stages multiply

(a)

Two mutually exclusive ways to make the journey:

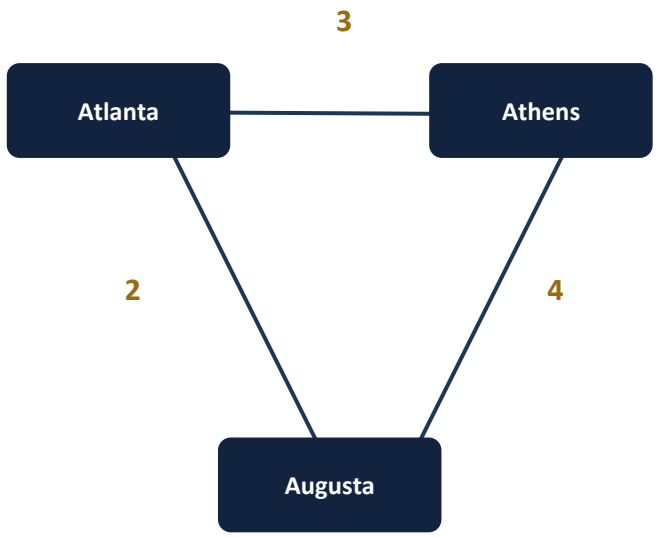
- via Athens — two stages, so multiply: $3 \times 4 = 12$
- direct — 2

The two options are alternatives, so add: $12 + 2 = 14$

(b)

Outbound and return are two stages, and every outbound choice leaves all 14 return choices available.

$14 \times 14 = 196$



$(3 \times 4) + 2 = 14$

ANSWER **(a) 14 ways (b) 196 round trips**

Q29 · Solution 2/2

Part (c) — when the stages are not independent

The return trip now depends on which routes the outbound used, so we split into cases.

Case 1: outbound went via Athens (12 ways)

One Atlanta–Athens route and one Athens–Augusta route are now used up.

Return via Athens: 3 remaining Augusta–Athens × 2 remaining Athens–Atlanta = 6

Return direct: 2

Returns available: 6 + 2 = 8, so this case contributes $12 \times 8 = 96$

Case 2: outbound was direct (2 ways)

One of the two direct routes is used up.

Return via Athens: $4 \times 3 = 12$. Return direct: 1 remaining. Total 13.

This case contributes $2 \times 13 = 26$

Combine

The two cases are disjoint, so add: $96 + 26 = 122$

ANSWER **122 round trips with no route repeated**

Q30 The product rule — filling boxes

PROBLEM

Count the possibilities in each design.

(a) A label identifier consists of one letter followed by three digits, repeats allowed.

(b) A bank password is two letters followed by two digits.

(c) A variable name is exactly 8 characters, each a letter or a digit, with the first required to be a letter.

Source · *Kolman, Counting §1, Example 1 and Exercises 1, 30*

Q30 · Solution

Draw the boxes, write the count under each, multiply

(a)

$26 \times 10 \times 10 \times 10 = 26{,}000$

(b)

$26 \times 26 \times 10 \times 10 = 67{,}600$

(c)

First position: letters only → 26 choices.

Each of the remaining 7 positions: 26 letters + 10 digits = 36 choices, repeats allowed.

$26 \times 36^7 = 26 \times 78{,}364{,}164{,}096 = 2{,}037{,}468{,}266{,}496$

Why repetition matters

Because repeats are allowed, every box keeps its full count. If characters had to be distinct, the counts would shrink at each step — that is the permutation case in Q32.



multiply across the row

one box per character position

ANSWER **(a) 26,000 (b) 67,600 (c) $26 \times 36^7 \approx 2.04 \times 10^{12}$**

Q31 Counting subsets

PROBLEM

Let S = {1, 2, …, 10}.

(a) How many subsets does a set with n elements have?

(b) How many subsets of S contain exactly 3 elements?

(c) How many subsets of S contain the element 1?

(d) How many contain neither 1 nor 2?

Source · *Kolman, Counting §1, Example 2. Reuse the bit-string picture from Q5.*

Q31 · Solution

Freeze what is constrained, free the rest

(a) 2^n — each element is independently in or out.

(b) Choosing a 3-element subset is choosing WHICH 3 of the 10 positions are in. Order does not matter, so this is C(10, 3) = 120.

(c) Force element 1 to be 'in'. The remaining 9 elements are unconstrained, each in or out: $2^9 = 512$.

(d) Force elements 1 and 2 to be 'out'. The remaining 8 are free: $2^8 = 256$.

The technique

Whenever a condition pins some elements, fix them and count the freedom that remains. Do not try to count around the constraint.



1 = in the subset, 0 = out

1 fixed, nine free bits → 2^9

ANSWER **(a) 2^n (b) 120 (c) 512 (d) 256**

Q32 Permutations — with and without repetition

PROBLEM

Compare these four questions carefully.

(a) How many 3-letter 'words' of DISTINCT letters can be formed from the letters of MAST?

(b) How many 3-letter 'words' can be formed from {a, b, y, z} if repeats ARE allowed?

(c) How many arrangements of all the letters of GROUP?

(d) Compute P(7,3), P(6,5), P(7,2).

Source · *Kolman, Counting §1, Examples 3, 8 and Theorems 3–4*

Q32 · Solution

One word in the question changes everything

(a) distinct letters — a permutation

$P(4,3) = 4!/(4-3)! = 4 \times 3 \times 2 = 24$

Boxes: $4 \times 3 \times 2$ — the count drops by one each time, because a used letter is gone.

(b) repeats allowed — not a permutation at all

$4^3 = 64$

Boxes: $4 \times 4 \times 4$ — the count never drops, because nothing is used up.

(c) all five letters, all distinct

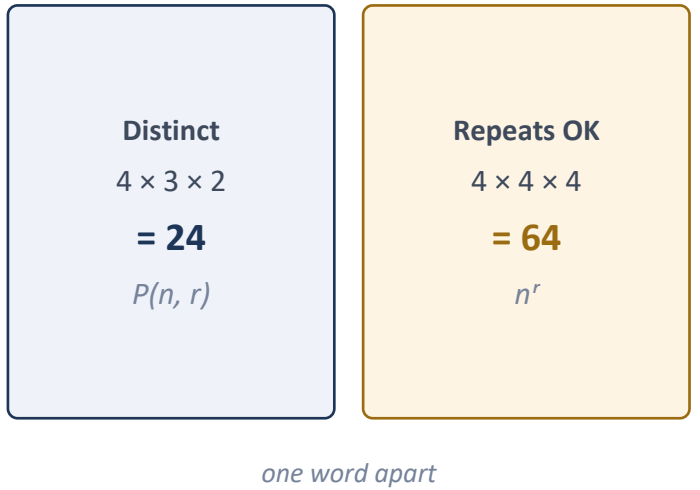
$5! = 120$

(d)

$P(7,3) = 7 \cdot 6 \cdot 5 = 210$ $P(6,5) = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 720$ $P(7,2) = 7 \cdot 6 = 42$

The comparison that matters

(a) and (b) are the same question except for the word 'distinct'. 24 versus 64. That single word is the whole difference.



ANSWER **(a) 24 (b) 64 (c) 120 (d) 210, 720, 42**

Q33 Permutations with constraints

PROBLEM

Arrangements where something is required.

(a) Six men and six women are seated in a row, anyone anywhere. How many arrangements?

(b) How many if men and women must alternate?

(c) How many arrangements of the letters of BOUGHT keep the two vowels next to each other?

Source · *Kolman, Counting §1, Exercises 17, 19. For (c), glue the vowels together.*

Q33 · Solution

The glue technique

(a)

12 distinct people in a row: 12! = 479,001,600

(b)

Only two patterns are possible: M W M W M W M W M W M W or W M W M W M W M W M W M.

Within a chosen pattern: 6! ways to place the men × 6! ways to place the women.

2 × 6! × 6! = 2 × 720 × 720 = 1,036,800

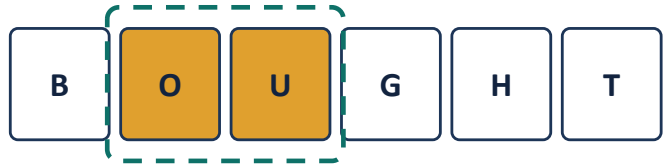
(c) BOUGHT — six distinct letters, vowels O and U

Glue O and U into a single block. We now arrange 5 objects: [OU], B, G, H, T.

5! = 120 arrangements of the five objects.

The block has 2! = 2 internal orders (OU and UO).

120 × 2 = 240



treat [OU] as ONE object → 5!
then × 2! for its internal order

5! × 2! = 240

ANSWER **(a) 479,001,600 (b) 1,036,800 (c) 240**

Q34 Permutations with repeated objects

PROBLEM

Find the number of distinguishable permutations of the letters in each word.

(a) BANANA (b) MISSISSIPPI

(c) PASCAL (d) ASSOCIATIVE

Source · *Kolman, Counting §1, Theorem 5 and Examples 9–10. Derive the formula, do not just apply it.*

Q34 · Solution

Tag the duplicates, count, then divide the tags out

The derivation, on BANANA

Temporarily tag the repeats: B, A_1, N_1, A_2, N_2, A_3 — six distinct symbols, 6! = 720 arrangements.

The two N's can be swapped without changing the visible word, so the 720 fall into pairs — divide by 2! .

The three A's can be permuted 3! = 6 ways without changing the word — divide by 3! .

720 / (2! · 3!) = 720 / 12 = 60

The general formula

For n objects where the types occur k_1, k_2, …, k_t times: $n! / (k_1! \, k_2! \ldots k_t!)$

Apply it

(b) MISSISSIPPI: 11 letters, M×1 I×4 S×4 P×2 → $11! / (1! \, 4! \, 4! \, 2!) = 34{,}650$

(c) PASCAL: 6 letters, A×2 → $6! / 2! = 360$

(d) ASSOCIATIVE: 11 letters, A×2 S×2 I×2, rest single → $11! / (2! \, 2! \, 2!) = 4{,}989{,}600$

ANSWER **(a) 60 (b) 34,650 (c) 360 (d) 4,989,600**

Q35 Circular permutations

PROBLEM

Seating around a round table.

(a) In how many ways can 7 people be seated around a circular table?

(b) In how many ways can n people be seated around a circular table?

(c) In how many ways if two particular people insist on sitting together?

Source · *Kolman, Counting §1, Exercises 24–26. Ask what makes two circular seatings 'the same'.*

Q35 · Solution

A circle has no first seat

Why (n−1)! and not n!

Rotating everyone one seat to the left produces a seating that is physically identical — the same people have the same neighbours.

So n! counts every genuine arrangement exactly n times, once per rotation.

n! / n = (n−1)!

Equivalent way to see it

Fix one person in a seat to serve as a reference point. This removes the rotational freedom. The remaining n−1 people are then arranged in a line relative to that person: (n−1)! ways.

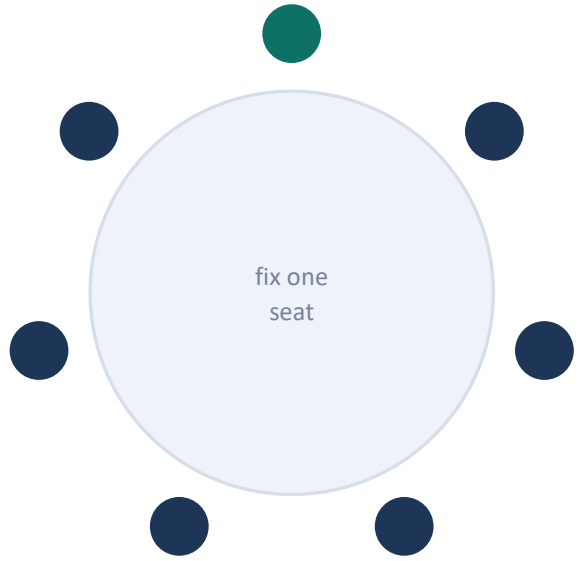
(a)

(7−1)! = 6! = 720

(c) two people must sit together

Glue the pair into one object. Now 6 objects sit around the circle: (6−1)! = 5! = 120.

The pair has 2 internal orders, so 120 × 2 = 240.



(7 − 1)! = 720

ANSWER **(a) 720 (b) (n−1)! (c) 240**

Q36 Combinations — when order stops mattering

PROBLEM

From a standard 52-card deck:

(a) How many distinct 5-card hands can be dealt?

(b) How many distinct ORDERED 5-card deals are there?

(c) Explain the ratio between the two answers.

(d) How many 6-card hands?

Source · *Kolman, Counting §2, Theorem 1 and Examples 2–3*

Q36 · Solution

P(n,r) = C(n,r) × r! — the whole relationship in one line

(a) a hand is a SET — order irrelevant

C(52, 5) = 52! / (5! · 47!) = 2,598,960

(b) a deal is a SEQUENCE — order recorded

P(52, 5) = 52 · 51 · 50 · 49 · 48 = 311,875,200

(c) the ratio

311,875,200 / 2,598,960 = 120 = 5!

Each unordered hand can be dealt in 5! = 120 different orders. That factor of r! is exactly the difference between the two ideas.

The derivation of C(n,r)

To build an ordered deal, first choose the set of r cards (C ways), then order them (r! ways). So C · r! = P(n,r), giving C(n,r) = n! / (r!(n−r)!).

(d)

C(52, 6) = 20,358,520

P(n, r) = C(n, r) × r!

choose the set, then order it

swap two chosen items —
different outcome?
YES → P NO → C

ANSWER **(a) 2,598,960 (b) 311,875,200 (c) ratio = 5! (d) 20,358,520**

Q37 Committees — choose within groups, multiply across

PROBLEM

Selection problems with independent groups.

(a) How many 7-person committees contain 3 women chosen from 20 available and 4 men chosen from 30 available?

(b) In how many ways can a committee of 3 faculty and 2 students be selected from 7 faculty and 8 students?

Source · *Kolman, Counting §2, Example 6 and Exercise 4*

Q37 · Solution

Within a group: combination. Across groups: product rule.

(a)

Task 1 — choose 3 women from 20: C(20, 3) = 1,140

Task 2 — choose 4 men from 30: C(30, 4) = 27,405

The two tasks are performed together, so multiply:

1,140 × 27,405 = 31,241,700

(b)

C(7, 3) × C(8, 2) = 35 × 28 = 980

The signal words

Within each group order is irrelevant — a committee is a set — so each group contributes a C.

The word 'and' joining the two requirements is the multiplication signal.

ANSWER **(a) 31,241,700 (b) 980**

Q38 'At least' and 'at most' — counting by complement

PROBLEM

An urn contains 15 balls: 8 red and 7 black. Five balls are chosen.

(a) In how many ways can all five be red?

(b) In how many ways can exactly two be red and three black?

(c) In how many ways can at most three be black?

(d) In how many ways can at least two be red?

Source · *Kolman, Counting §2, Exercises 10–13. For (c) and (d), consider counting what you do NOT want.*

Q38 · Solution

Subtract the unwanted cases

Total selections of 5 from 15: $C(15, 5) = 3{,}003$

(a) $C(8, 5) = 56$

(b) $C(8, 2) \times C(7, 3) = 28 \times 35 = 980$

(c) at most three black — the direct route is four cases

0B: $C(7,0)C(8,5) = 56$ 1B: $C(7,1)C(8,4) = 490$ 2B: $C(7,2)C(8,3) = 1{,}176$ 3B: $C(7,3)C(8,2) = 980$

Sum = 2,702

(c) by complement — two cases instead of four

Unwanted: 4 or 5 black. $C(7,4)C(8,1) = 280$ and $C(7,5)C(8,0) = 21$

$3{,}003 - (280 + 21) = 3{,}003 - 301 = 2{,}702$ ✓

(d) at least two red

Unwanted: 0 or 1 red. $C(8,0)C(7,5) = 21$ and $C(8,1)C(7,4) = 280$

$3{,}003 - 301 = 2{,}702$

ANSWER **(a) 56 (b) 980 (c) 2,702 (d) 2,702 — the same event, described differently**

Q39 Selection with repetition — stars and bars

PROBLEM

Choosing when repeats are allowed and order does not matter.

(a) A radio station lets a winner choose 3 CDs from the Top Ten list. Repeats are allowed and order is irrelevant. In how many ways?

(b) In how many ways can 10 identical chocolates be distributed among 4 children?

Source · *Kolman, Counting §2, Theorem 2 and Example 4. The formula is C(n+k−1, k).*

Q39 · Solution

Turn the problem into arranging stars and bars

The model

Represent the k chosen items as stars and use n−1 bars to divide them into n categories.

Example for (b): ★★ | ★★★★ | ★ | ★★★ means child 1 gets 2, child 2 gets 4, child 3 gets 1, child 4 gets 3.

Every arrangement of stars and bars is one valid distribution, and every distribution gives one arrangement.

So we are just choosing which of the (k + n − 1) positions hold the bars — a combination.

(a) n = 10 CDs, k = 3 choices

C(10 + 3 − 1, 3) = C(12, 3) = 220

(b) n = 4 children, k = 10 chocolates

C(4 + 10 − 1, 10) = C(13, 10) = C(13, 3) = 286



10 stars + 3 bars = 13 slots
choose which 3 are bars → C(13, 3)

2, 4, 1, 3 chocolates

ANSWER **(a) 220 ways (b) 286 ways**

Q40 Mixing the rules

PROBLEM

A valid password consists of exactly seven characters. The first must be a letter chosen from {A, B, C, D, E, F, G}; each of the remaining six may be any of the 26 English letters or any of the 10 digits, with repeats allowed. How many passwords are possible?

Source · *Kolman, Counting §2, Example 5. Two stages — identify them before computing anything.*

Q40 · Solution

Decompose into stages, then multiply

Stage 1 — the first character

Chosen from a 7-element set: 7 ways.

Stage 2 — the remaining six characters

Each position has 26 + 10 = 36 options, and repeats are allowed, so positions are independent.

36^6 = 2,176,782,336 ways.

Combine

The two stages are performed in sequence, so multiply:

7 × 2,176,782,336 = 15,237,476,352

Why 36^6 and not P(36,6)

Repeats are permitted, so using a character does not remove it from the pool. The count stays at 36 for every position.



multiply across the row

restricted position first

ANSWER **15,237,476,352 possible passwords**

Q41 Counting with inclusion–exclusion

PROBLEM

Among the integers from 1 to 1000 inclusive:

(a) How many are divisible by 3 or by 5?

(b) How many are divisible by neither?

Source · *This is the set-theory principle from Q7–Q9 doing service as a counting tool.*

Q41 · Solution

The inclusion–exclusion principle, reused

Let A = multiples of 3 in the range, B = multiples of 5.

$|A| = \lfloor 1000/3 \rfloor = 333$ — *every third integer*
$|B| = \lfloor 1000/5 \rfloor = 200$ — *every fifth integer*
$A \cap B$ = multiples of 15, since lcm(3,5) = 15

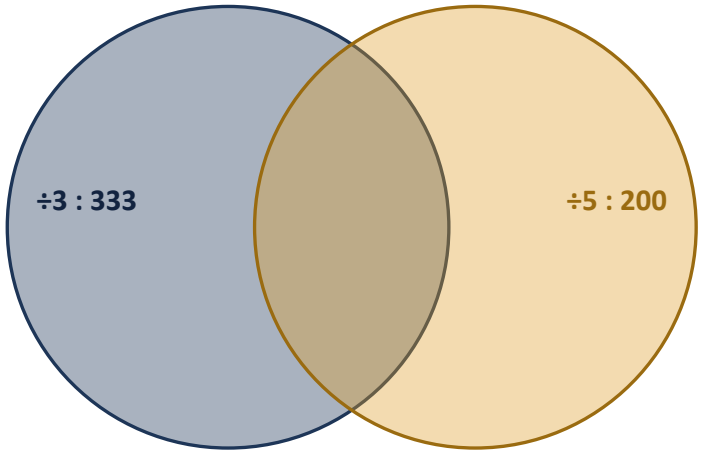
$|A \cap B| = \lfloor 1000/15 \rfloor = 66$

(a)

$|A \cup B| = 333 + 200 - 66 = 467$

(b)

Neither = $1000 - |A \cup B| = 1000 - 467 = 533$



66 in the overlap

ANSWER **(a) 467 integers (b) 533 integers**

Q42 Bit strings — counting in the language of computing

PROBLEM

Consider bit strings of length 8.

(a) How many are there in total?

(b) How many contain exactly three 1s?

(c) How many begin with 1 and end with 00?

(d) How many contain at least one 1?

Source · *Every part of this problem is a technique from an earlier slide, in computing costume.*

Q42 · Solution

Every counting idea in this unit, on one object

(a) product rule

8 positions, 2 choices each: $2^8 = 256$

(b) combination — choose which positions hold the 1s

C(8, 3) = 56

A length-8 bit string with three 1s IS a 3-element subset of {1,…8}. Same object, two names — exactly the point made in Q31.

(c) freeze the constrained positions

Position 1 is fixed at 1; positions 7 and 8 are fixed at 0. That leaves 5 free positions.

$2^5 = 32$

(d) complement

'At least one 1' is everything except the single all-zeros string.

$2^8 - 1 = 255$



three 1s ↔ the subset {1, 3, 4}

C(8,3) = 56 ways to place them

ANSWER **(a) 256 (b) 56 (c) 32 (d) 255**

Quick reference — set theory & logic

Set identities

De Morgan: $(A \cup B)' = A' \cap B'$, $(A \cap B)' = A' \cup B'$

Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Difference: $A - B = A \cap B'$

Symmetric difference: $A \oplus B = (A - B) \cup (B - A)$

Cardinality

$|A \cup B| = |A| + |B| - |A \cap B|$

$|A \cup B \cup C| = \Sigma|A| - \Sigma|\text{pairs}| + |A \cap B \cap C|$

$|P(A)| = 2^n$ where $n = |A|$

$|A \times B| = |A| \cdot |B|$

Proof techniques

Direct: assume p, derive q

Contrapositive: assume ¬q, derive ¬p

Contradiction: assume $p \wedge \neg q$, derive an absurdity

Induction: basis $P(n_0)$, then $P(k) \Rightarrow P(k+1)$

Logical equivalences

$p \Rightarrow q \equiv \neg p \vee q$

$\neg(p \Rightarrow q) \equiv p \wedge \neg q$

$p \Rightarrow q \equiv \neg q \Rightarrow \neg p$ (contrapositive)

$\neg(p \wedge q) \equiv \neg p \vee \neg q$, $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Quick reference — counting

The two rules

Sum rule: disjoint alternatives → add
Product rule: sequential stages → multiply
The sum rule requires the cases to be disjoint
Product rule needs stage 2's count to be the same whichever option was taken at stage 1

The four selection cases

order matters, no repeats: $P(n, r) = n!/(n-r)!$
order matters, repeats: n^r
order irrelevant, no repeats: $C(n, r) = n!/(r!(n-r)!)$
order irrelevant, repeats: $C(n+k-1, k)$

Useful identities

$P(n, r) = C(n, r) \times r!$
$C(n, r) = C(n, n-r)$
Distinguishable permutations: $n! / (k_1!\, k_2! \ldots k_t!)$
Circular arrangements: $(n - 1)!$

Techniques worth naming

Glue — items that must stay together become one object
Freeze and free — pin the constrained, count the rest
Complement — count the unwanted, subtract from the total
Stars and bars — identical items into distinct bins

Where this goes next

Sets → relations, functions, and equivalence classes — all built on subsets of a product

Logic → predicates and quantifiers, then formal verification and Boolean circuit design

Counting → the pigeonhole principle, the binomial theorem, and recurrence relations

Kolman, Busby & Ross · Discrete Mathematical Structures, 6th edition